

An In-Depth Analysis on Incorporating Autonomous Systems

*Exploring the Road Ahead: A Deep Dive into Autonomous Systems

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Abstract—This review report brings forth certain aspects that current industrial megalomaniacs are facing, in the realisation of carving out error-free Autonomous Vehicles(AVs). Having said so, the report also spells out a few influential factors that affect the practical applications of these automated systems. The goal behind incorporating such a vague document is to focus on few factors which are usually ignored while idealizing their applications. The document will not only address current technological readiness of consumers but also point out few commercialized applications which tend to be ignored due to lack of feasibility. Furthermore, this report signifies a possibility of co-existence of AV technologies with human-driven technologies(HVs), for a substantial time providing dedicated lanes and deducting network performances in urban areas. Next, the report scrutinizes the regulatory discrepancies surrounding AV deployment, current framework and improvement areas that the governing bodies are in deficit of. Along with this, the report summarizes the control techniques, intricate electronic architecture, safety impacts and future directions of the AVs in a nutshell.

Index Terms—megalomaniacs, influential factors, technological readiness, co-existence, dedicated lanes, electronic architecture

I. INTRODUCTION

Since the dawn of Industrial Revolution, mass production has been a pivotal goal for industrialists to conquer and has created a rat-race among many to accomplish it. The competitiveness of this juncture has increased ten-folds and has resulted reducing the manufacturing lead time as low as possible. This shifted the manual-labour based economy to machine-based economy, applicable in agriculture, manufacturing, production and transportation. Adding on, the Information age of this era, the automation capabilities of mechanisms have sky-rocketed along with solving complex problems within seconds. Factoring in the Intelligence Quotient of average humans, handling complex mechanisms is now withholding the potential the particular autonomous intelligent mechanism can achieve. This is why we see a slow transformation of a new trend, where automation is being replaced by autonomy.

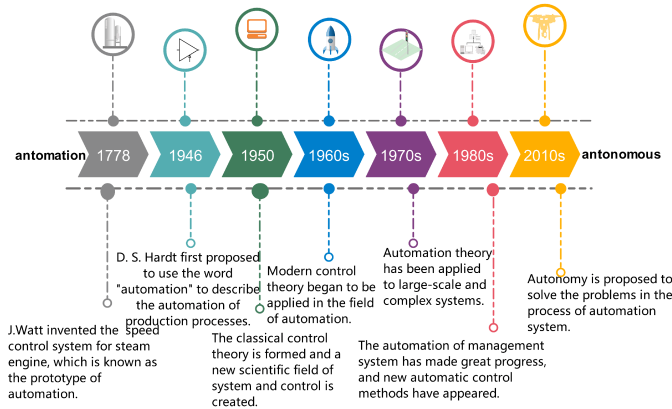
Examples of this inference are many, such as Subway Systems in some cities working without humans efficiently, autonomous logistic cargo systems, autonomous wheelchairs, Varied Systems in International Space Station, agents in the internet environment, etc.

An industrial autonomous system is a delimited technical system, which systematically and without external intervention, achieves its set objectives despite uncertain environmental conditions. This definition captures four essential characteristics-

- systematic process execution
- adaptability to uncertain environmental conditions
- self-governance
- self-containedness of the system. [1]

On the other hand, the social perception of accepting the autonomous systems can be keenly observed in the society, as advanced technical systems are taking over more and more tasks in our society. This has resulted the transformation of machines from being a mere tool to a type of companion or teammate. Now these machines can't be categorised as decision supporters but decision makers. Moreover, the implementation of increasingly autonomous technical systems is associated with a wide array of social questions. These questions are determined through actions of autonomous technical agents which are becoming more intractable, opaque and difficult to comprehend. This calls for exploring the extent to which increasingly autonomous systems are seen as responsible for their actions and whether decisions made by the autonomous algorithms are seen as trustworthy or fair. All in all, what matters the most here is how people subjectively perceive technical systems and their actions regardless of the capability of the machine in itself. Even though this conundrum, being emphasised regularly and yet to be solved completely, is hard to predict how increasing levels of autonomy would shape such perceptions. [2]

For an instance, if the automobile industry is to be considered, more and more new vehicles are incorporating driver assisting features making them partly autonomous vehicles. Additional features with the help of sensors and on-board cameras have given rise to self-driving cars. Furthermore, it now seems very much possible the development of autonomous vehicles, where it can navigate from a source to a destination by using various technologies totally free of human intervention. [3]From a social perspective, society is front-



[Source: [4]]

Fig. 1. Evolution of Autonomous Systems

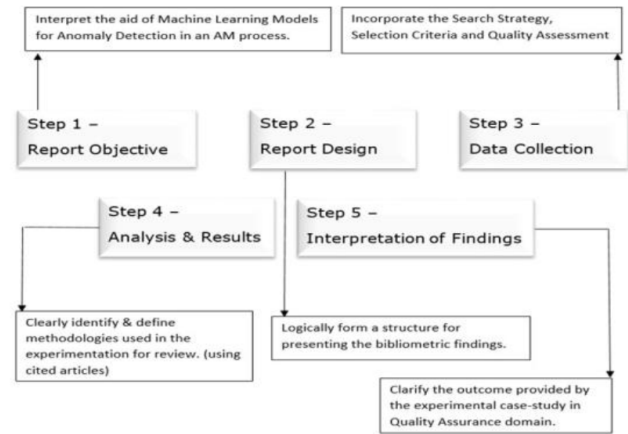


Fig. 2. Roadmap

runner in accommodating these type of autonomous systems as compared to other autonomous technologies.

II. LITERATURE REVIEW

A. LITERATURE SEARCH STRATEGY

The search was carried out from the assigned date till 1st of March. The three databases IEEE Explore, ASME, EBSCO Host and Research Gate were considered. Since the paper title was established from a Case- Study, it was bifurcated in three parts, viz., Autonomous Systems, Social Impact and Incorporation of the System. Correspondingly, the search term combination: title or abstract or keywords containing these technical parameters were considered. The search resulted in articles that met the inclusion criteria. Furthermore, the search provided extensive results, which were then filtered out to display only journals, reports and published conference papers.

B. SELECTION CRITERIA FOR STUDY

Parsing through the resultants, only correlating concepts to the discerned title were considered for writing the report. Also, the papers which categorized under international peer-reviewed publications were the only ones taken in consideration, to guarantee high quality standards. After reviewing 40 articles, only 17 articles were selected as relevant to the topic as they deal with Socio-Demographic relations of Autonomous Systems with the people who would ultimately use the systems. Due to the segregation of the title in subtitles and then correlation of those papers with each other on the basis of the whole title, increased the number of research papers with respect to the selection criteria. The papers with similar concept but varying in terms of applications were excluded.

C. ASSESSING QUALITY OF PAPERS

The quality of papers is assessed using different criteria's considered to ensure rigor, relevance and contribution to the field. The Abstract provided in each research paper was usually proven to be the detrimental factor, followed by the form of objective addressable. The 'Literature Review' presented

in those papers were scrutinized for gaps in the current literature that this case-study aimed in addressing. Lastly, the contribution of new insights and experiments were examined on how they would advance the knowledge in the field. Also a neat appropriation of citations and references was checked, so as to understand the author's perception for the proposed experimentation.

D. Bibliometric Analysis

III. FACTORS AFFECTING THE ADOPTION

A. Behavioral Reasoning and Technology Readiness

The Behavioral Reasoning Theory(BRT) is an exemplary theoretical model whenever a new process is to be introduced in the market. It determines examining of relative effects of resistance and adoption of innovation. The theory not only distinguishes among "reasons for" and "reasons against", but also assists in assessing the impact of consumer showcased attributes on actions utilizing a single decision-making framework. Hence, BRT for the domain of Autonomous Systems has helped in exploring how psychological traits and reasoning process, according to BRT framework, influences consumer adoption and attitudes.

However, research using BRT framework is limited in terms of individual and personality characteristics. This is when Technological Readiness(TR) comes into the picture. TR is nothing but the psychological state shaped by various mental factors that influence an individual's inclination to embrace new technology. [5] Not only TR but the BRT framework readily embraces other such parameters viz.-

- Personal Optimism.
- Consumer's Technological Innovativeness.
- Perceived Discomfort when in application.
- Perceived insecurity toward new technology.
- Social Influence.

B. Trustworthiness

As technology continues to advance, researchers have developed autonomous systems that pose a potential threat to users

due to their ability to overpower human control. Therefore, ensuring trustworthiness has become crucial to maintain optimal system performance. Establishing trust in autonomous systems requires an understanding of trust in human-to-human relationships and interactions between humans and autonomous systems. Trust can be conceptualized using a two-factor model that combines ability and ethics, encompassing factors such as competence, reliability, integrity, motives, and character.

One challenge in building trust in autonomous systems is that much of the data and algorithms used by these systems are hidden from most human stakeholders, making it difficult for them to understand even with explanations provided in plain language. This raises the issue of implicit and explicit justifications of trust. Implicit trust develops without conscious evaluation, relying on pre-conscious, embodied responses and reputational features of the system's designers. Explicit trust, on the other hand, is based on deductive, inductive, or abductive reasoning from the behavior and testimony of the agent.

Implicit indicators of trust may manifest in physical responses and intuitions, while explicit trust relies on reflective judgments and rational assessments based on past experiences with similar systems. As autonomous systems become more competent, trust in them tends to increase, both implicitly and explicitly, until their competence surpasses human comprehension. However, establishing explicit trust may remain elusive, requiring ongoing evaluation of the system's performance and reliability over time. [6]

C. Specimen Exemplification

To better understand the growing concerns of accepting AS in the current socio-economic trends, Shared Autonomous Vehicles(SAVs) stand out to be a prime example. Since these are crucial elements of sustainable future mobility concepts. In this ever-increasing urbanization, with the help of advanced technologies and rising shared economies, smart city structures will be shaped acutely in the coming future. Upon expressive consumer surveys, feedbacks and studies certain factors related to different paradigms have culminated. These factors can be categorised into three main sections.

- 1) **User Related Factors** - These contain **Socio-Demographic Factors**, such as, Age, Gender, Household Income, Educational Level, Employment Stats, Residence Location, Children in Household, Mobility Impairments, Household Size and Working/Living in Pilot Area. Moreover **Mobility Behaviour Related Factors** include Vehicle Ownership, PT User/Experience, Driving License, Private Vehicle User, Sharing User Experience, SAV Experience, Trip Purpose, Crash Involvement, Ease of Parking, Driving Experience and Daily Travel Time. Furthermore, **Socio-Psychological Factors** also affect the SAV implementation viz. Perceived Ease, Social Influence, Perceived Usefulness, Performance Expectations, Trust, Perceived Risks, Privacy Concerns, Perceived Benefits, Hedonic Motivation and Habit. Lastly, **Attitudinal Factors** are Awareness towards AVs, Personal Innovativeness, Interest in Sharing,

Environmental Concern, Driving Passion, Cost-Oriented User and Health.

- 2) **Service Related Factors** - These factors constitutes the following - Travel Cost, Safety and Security, Accessibility, Comfort, Service Attributes and Quality, Additional Passengers, Vehicle Features and Design, Presence of Staff/Supervision, Reliability and Facilitating Conditions.
- 3) **Time and Weather - Related Factors** - These factors are Travel Time, Waiting Time, Value of Time, Travel Speed and Weather Condition. [7]

D. Implications

Autonomous Systems on its own means an entity that can perform various tasks and procedures on its own without any sort of human intervention. Implementation of such systems will bring upon implications in the following manner :

- **Implementation Pathway:** - International Transport Forum(ITF) in 2015, outlined deployment pathways for autonomous vehicles based on vehicle type and level of automation.

First and Foremost starting with Trucks and Light-Duty Vehicles on Freeways are likely to start functioning due to more uniform design, organized traffic flows, high speeds and predictable conditions.

Secondly, Passenger and Delivery Autonomous Transport where initial deployment can take place with low-speed operation in restricted areas such as campuses, resorts and airports.

Thirdly, Urban Mobility Autonomous Vehicles will harbour evolution including increased speed and decreased restrictions on deployment area and infrastructure, which will eventually operate in mixed traffic.

Fourthly, Autonomous Private Vehicles will need features added at each level , including Automated Parking System, Traffic Jam Assistance, Highway and Traffic Jam Chauffers, Highway Pilot and Parking Garage Pilot.

Fifthly, Autonomous Trucks will include vehicle platooning for fuel savings, alongwith Cooperative Adaptive Cruise Control platooning and full Truck platooning.

- **Benefits vs Negative Effects:** - The impact of autonomous systems especially for vehicles on society and lifestyles vary depending on their implementations. Of which, a few Potential Benefits offered by these systems are Safety, Cost Reduction, Traffic Reduction, Accessibility, Equitable Land Use, Environmental Benefits and Productive Usage of time. Whereas, the potential negative effects can be pointed out as Increased Vehicle Ownership, Job Displacement, Urban Sprawl, Ethical Dilemmas, Safety Concerns.
- **Safety:** - By doing away with human error, which causes over 90 percent of collisions, the move to fully automated cars has the potential to significantly lower crash rates. Not only may this increase in safety, prevent fatalities, but it could also lessen unintended consequences including medical bills, legal fees, traffic jams, and infrastructure

damage. Nonetheless, autonomous vehicle-related traffic incidents have resulted from present flaws in automated systems. There may be worries about safety with autonomous public transportation vehicles, particularly for particular demographics like female riders. While autonomous subway trains and other high-capacity vehicles are widely regarded as safe, smaller autonomous vehicles that carry fewer passengers may encounter difficulties guaranteeing passenger safety and fostering user confidence. This can discourage users from making use of these services.

- **Vehicle-Kilometer Travelled:** - Vehicle kilometers traveled (VKT) will be significantly impacted by the deployment of autonomous cars, which will depend on a number of variables including legislation, pricing, and deployment forms. Although autonomous cars provide advantages like less parking and fuel expenses, they can also raise the value of a vehicle due to induced demand from underserved user groups, the greater allure of driving, and trip reallocation.
- **Congestion:** - Three major variables pertaining to autonomous cars impact traffic congestion:
 Decrease in Accidents: Decreased delays and traffic congestion result from fewer accidents, especially when it comes to lane closures brought on by collisions. This raises the overall reliability of the transport system.
 Optimized Throughput: Because autonomous cars provide smoother rides, have greater speed limits, and require shorter following distances between other cars, they may be able to expand the capacity of the road.
 Variations in VKT: Increased traffic may result from the predicted increase in Vehicle Kilometers Traveled (VKT) brought on by elements such as stimulated demand.
 Congestion pricing can efficiently control traffic flow, particularly when combined with cutting-edge communication systems. Furthermore, communication between autonomous cars has the potential to replace traffic signals and drastically cut down on delays, especially at crossings. According to estimates, vehicle-to-vehicle communication might result in a 445 percent increase in road capacity and safer, more efficient travel.
- **Value of Time and Accessibility:** - Time can be saved by the driver using autonomous vehicles to shift their focus from driving to other duties. Since they can continue working while on board, business travelers may find this to be really helpful. By lowering the number of transfers, walking distance to and from stops, and waiting times, autonomous cars deployed for on-demand door-to-door or street-corner-to-street-corner public transportation can also contribute to increased accessibility and connectivity. Particular improvements can be made to accessibility for young, old, and disabled passengers who now use traditional public transportation but do not possess a driver's license.
- **Costs:** - It is anticipated that the widespread use of autonomous vehicles will result in cost savings from both

internal and external sources. Savings will directly result from less gasoline being used, lower insurance and maintenance costs for vehicles as a result of increased safety, and lower labor costs for taxi and public transportation services. There will also be an increase in employment prospects for infrastructure and driverless cars. From the outside, it is expected that autonomous cars will ease problems like traffic and pollution by improving accessibility, making parking spots available, alleviating traffic, and building up road capacity. Nevertheless, depending on usage habits, the effect on externalities can change. By maximizing vehicle utilization and lowering total Vehicle Kilometers Traveled, the deployment of autonomous vehicles in shared mobility schemes can minimize negative externalities (VKT). problems with the environment and traffic.

- **Cybersecurity and Privacy:** - Cybersecurity issues are associated with the use of advanced information technology to link and control autonomous vehicles. Potential vulnerabilities include GNSS spoofing and the introduction of fraudulent messages into cooperating automated vehicles. These assaults may cause vehicles to behave erratically and put lives in danger. Security guidelines must be created, and extra software and misbehavior detection systems must be deployed on cars and associated infrastructure in order to reduce these dangers. Integrating data from several sources can help detect security risks. However, gathering information for communications and vehicle control, as well as for system security and safety, presents privacy issues. Vehicles and digital infrastructure should be built with privacy in mind to address this. Technical safeguards against the gathering, storing, or sharing of certain personal data may be necessary for this, as well as the incorporation of data encryption and anonymization methods. It is crucial to develop relevant legal and technical standards. [8]

IV. ASPECTS OF AUTONOMOUS SYSTEMS

Autonomous Systems consists of various components that participate in intelligently carrying out desired activities efficiently. Few of which components are presented below as:

- 1) **Agents:** - These are physical entities which act upon receiving some kind of input depending upon the perceptions made.
- 2) **Sensors:** - These are devices which help the agent to perceive and understand the environment based on the gathered data.
- 3) **Actuators:** - These are mechanisms driven by the agent to realise the decisions made.
- 4) **Decision-Making Algorithms:** - These are software procedures trained by the user on the agent to draw conclusions.
- 5) **Communication Infrastructure:** - This comprises of networks, protocols and interfaces that transmit information between agents and external system.

- 6) Control System: - These systems supervise the decisions made by the agent to regulate them according to algorithms.
- 7) Learning Mechanisms: - These entities are introduced in the system, so as to improve performance over time through experience and feedback.
- 8) Safety Mechanism: - These look after error detection, fault tolerance and fail-safe behaviours.
- 9) Localization Mapping: - This helps the system in determining self position and build environment representations for dynamic environments.

Now, let's delve into the concept of autonomy, what it exactly means and how it is implemented on a system. The best way to understand autonomy is to draw parallels between the developmental process of biological agents and design principles of artificial agents. Upon which autonomy emerges from the ability to master tasks necessary for survival in a given environment. The characteristics of environments that influence the tasks agents undertake can be singled out one-by one as:

- Observability
- Determinism
- Episodic Nature
- Static vs. Dynamic
- Discrete vs. Continuous
- Single vs. Multiagent

To ensure effective autonomy, clear conception of an agent's habitat with its range of environment is essential. Not only the environmental fit but the role of intelligence enabling agents in adapting to their specific environment is also noteworthy. Since higher intelligence widens the range of adaptability to harsh environments. In conclusion, the importance of understanding and leveraging the interplay between agents and their environments to achieve autonomy is of the paramount.

Humankind since the dawn of early age, is curious about human anatomy and cognitive understanding which has driven the motivation of finding how we think, act and behave. We can be called as a primordial candidate of being classified into Intelligent Autonomous Agents(IAAs). Similar to which, motivation can be focused on two main drives: i) understanding human behaviour and ii) improving current technology.

The first motive is a long-standing desire to understand the intricacies of human cognition, from primal reflexes to more advanced cognitive abilities like consciousness and problem-solving. This endeavor, which has its roots in philosophy and psychology, has gained traction in science thanks to the development of fields like cognitive science and artificial intelligence (AI). These domains have given rise to a methodological framework for computational research on behavior like that of humans. The second reason has to do with the growing intricacy of technical systems, such as personal computers (PCs). Even with advances in high-level programming languages and functional design, consumers still frequently find it difficult to navigate many features of current PCs. The need for IAAs—intelligent installation assistants—that

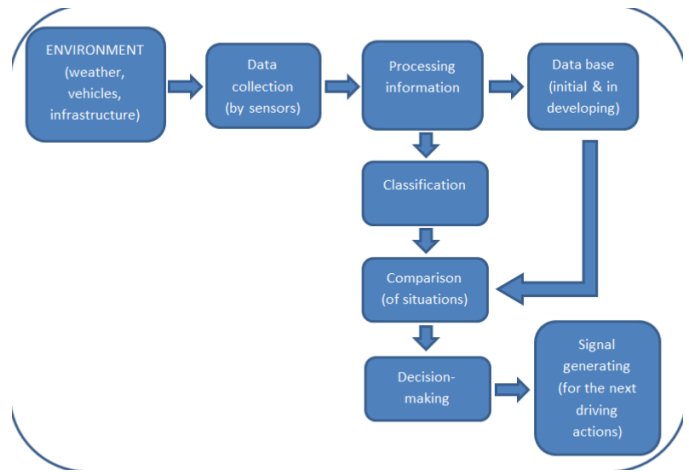


Fig. 3. Process of AS [Source: [10]]

can handle jobs on their own without assistance from users has arisen as a result of this complexity.

The adoption of IAAs though offer potential solutions for managing complexity, reduce labour costs and enhance efficiency. It brings about about new challenges and risks such as ethical, legal, political and psychological implications regarding accountability, regulation and human interactions with IAAs. To name a few challenges related to IAAs -

- 1) Uncover the mystery of consciousness and its role in IAAs.
- 2) Bridge the gulf between high-level and low-level functions in IAAs.
- 3) Bridge the gulf between high level cognitive functions and the functions realizable by metaheuristics.
- 4) Implant truly high-level intelligence functions into IAAs.
- 5) Develop the logicist approach to realizing AI. [9]

The Figure depicted above shows how any action of any autonomous system will follow through in reacting as required. The section addresses autonomous systems and how they interact with the real environment; it does not cover software-only structures like agents or viruses. While acknowledging the parallels between software agents and physical autonomous systems, it highlights the particular difficulty of interacting and moving in the open physical world in order to accomplish challenging objectives. Autonomy already is well-established in several relevant application domains making use of current capabilities and particular science and technology challenges. The domain could have systems operating in maritime, ground, air and space. These domain specific aspects include sensing and perception, manipulation, mobility, power, navigation and communications. While it is impossible to fully separate them from autonomy, we are concentrating on those features of it that might be consistent across different operational domains. These consist of behavior coordination, model-based and reactive control, layered control, and automated planning.

A. Planning and Scheduling

The first step towards autonomous behavior is the creation of a strategy to achieve a desired objective or collection of goals within the limits and available resources. Artificial intelligence has made progress in automating planning processes, as demonstrated by the development of hierarchical task decomposition and state-space search. Planning capacities have been further improved by recent methods, such as Markov decision processes, which have found effective applications in a variety of fields, including manufacturing process management, transportation, and military logistics.

B. Layered Control

Traditionally, a large portion of research in the field of automated planning has treated the problem as an offline process, with the aim of creating an optimal plan given an initial state, a goal state, and a set of actions. This method, however, is inadequate for autonomous systems working in open environments, because not all aspects are fully described. Planning must take place in tandem with plan execution in these kinds of situations in order to properly address unforeseen circumstances. Planning in open environments necessitates constant adaptation and adjustment to handle unforeseen variables as they happen, in contrast to the offline method.

The challenge of attaining efficient control in autonomous systems is tackled by the canonical design, which is built on stacked control loops. This design seeks to provide responsive and reliable control by dividing control mechanisms into layers that function at various timescales and levels of abstraction. By guaranteeing that autonomous systems can adapt and respond correctly to changing situations and inputs while maintaining steady performance, it addresses the challenges associated with controlling autonomous systems. From undersea vehicles to exploration spacecraft, autonomy systems using this design are now practically universal. Even a more complex version has been suggested as the industry standard for intelligent manufacturing systems.

C. Model Based Reasoning

One can face several complexities of planning and control in autonomous systems, these complexities highlight various dimensions of behavior tailored to specific autonomy domains. Such as Navigation, Mobility, Power Management, Health Management and Pay-load specific operations. It becomes quite hectic to integrate these functional areas due to inherent coupling of subsystems and shared resources. Analytical models are the foundation of classical control theory, but as systems get increasingly complicated, it might not be possible to build a single, all-encompassing model. As a result, heuristic methods based on control rules have been developed; however, the combinatorial explosion of subsystem interactions may pose limitations to them.

Complex autonomous system development is aided by the formal validation of autonomous control system performance, which is based on explicit behavior models. The current methods are a break from traditional control theory and stress

the use of explicit system models for efficient autonomy design, even if they borrow upon early work in digital circuit diagnosis.

D. Reactive Behavior

Reactive control runs quickly and has deterministic limitations; it is in charge of preserving local system stability. It includes things like vehicle "safing," which makes ensuring that system telemetry stays within predetermined parameters and, if needed, initiates quick corrective action. Emergent behaviors were identified by early closed-loop control studies, which prompted Rodney Brooks to develop the "subsumption" design in the mid-1980s. Rather than depending on global system models, this method achieves complicated behaviors by stacking and mixing locally optimum controllers. Subsumption architecture is limited in its ability to implement complicated behaviors, even while it makes model specification and computation simpler. Reactive autonomous controllers are finding use in many different sectors and are becoming more and more popular for coordinating autonomous activities in spite of these drawbacks.

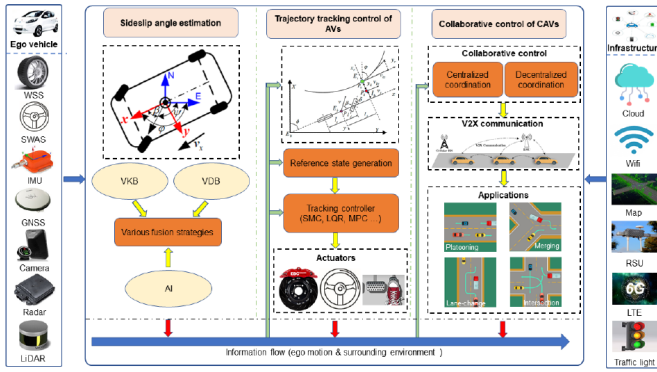
E. Behavior Coordination

Coordinated behaviors are needed in many sectors, from satellite constellations to underwater gliders, which makes developing autonomous systems extremely difficult. There are numerous avenues of research into coordinated autonomous behavior. Research in the field of software agents focuses on coordination languages, knowledge representation, and communication infrastructure. Provable system characteristics are the focus of distributed control research, which tackles functional capabilities such as relative motion control for vehicle formations. Research on distributed autonomy is dominated by two main techniques: model-based and reactive methods. Model-based strategies seek to reduce communication in distributed controllers while preserving global state information. However, in multi-vehicle systems, scale problems are common. Swarm intelligence-inspired reactive techniques mimic social insect behavior in the absence of clear coordination schemes or worldwide communications.

Swarm intelligence has shown early success, which sparked ongoing efforts to use it in distributed autonomy applications such as formation flying, search, and pursuit. Even if they show promise, solely reactive approaches can't always achieve complicated behaviors. One of the most important research challenges to enable sustained behavioral synchronization and increased operational complexity in autonomous systems is to integrate model-based deliberative coordination with reactive techniques. [11]

V. CONTROL TECHNIQUES

As an exemplification of Autonomous Systems, the already implemented control techniques of Autonomous Vehicles can be considered for further research purposes. When developing an autonomous vehicle, its state information is of utmost importance, for an example estimation of sideslip angle,



[Source : [12]]

Fig. 4. Vehicle Control Framework

longitudinal and lateral velocity, orientation and tire forces, etc. This is crucial for various aspects of vehicle control in autonomous and connected autonomous vehicles. Although difficult, sideslip angle estimate is necessary for many applications, including motion planning, handover modules, road condition estimation, vehicle stabilization, and navigation.

Moving beyond state estimation, next technique to control is the vehicle trajectory tracking control, which involves various control strategies such as Proportional-Integral-Derivative(PID) control, Linear Quadratic Regulator(LQR), feed-forward and feedback control, robust control, Sliding Mode Control(SMC), Model Predictive Control(MPC) and learning based methods. Even with the advancements in model-based control algorithms, problems still arise, particularly under extreme driving situations. This has prompted research into learning-based approaches because of their flexibility and capacity for self-optimization.

An easy amalgamation of these control strategies is the collaborative driving automation in intelligent transportation system(ITS), emphasizing the importance of Connected Autonomous Vehicles(CAVs). One important component of CAV development is collaborative control, which includes mixed, decentralized, and centralized control strategies, each having pros and cons. To improve the effectiveness and safety of CAVs, current research concentrates on particular driving scenarios such as platooning, lane changes, merging, and intersection management.

A. Vehicle State Estimation

The motion planning, decision-making and control techniques of AVs and CAVs are conceived from the critical vehicle state information, including position, velocity(both longitudinal and lateral) and attitudes(roll, pitch and yaw). For estimating a sideslip angle crucial for Navigation Systems it is also significant for heading angle estimation. Another point to note is that Kalman Filters and nonlinear observers are dominant methods for vehicle state estimation. The slideslip angle estimation approaches are presented below as:

- Onboard-Sensor-Based(OSB) approach - It relies on data from sensors directly installed on the vehicle. This

approach encompasses three main subcategories viz. Vehicle-kinematic-Based approach, Vehicle-dynamic-based approach and Dynamic-kinematics-fusion-based(DKFB). OSB approaches leverage sensors such as wheel speed sensors, steering wheel angle sensors, inertial measurement units (IMUs), and global navigation satellite systems (GNSS) to estimate the sideslip angle, making them suitable for practical implementation in mass-produced vehicles.

- GNSS-augmented (GAU) approach - The global navigation satellite systems (GNSS) are used to enhance onboard sensor data with information in the GNSS-augmented (GAU) approach. To estimate the sideslip angle, this method usually integrates vehicle dynamics models with GNSS location observations. The availability of GNSS data, which offers global positioning information, is advantageous to GAU approaches as it improves the accuracy and dependability of sideslip angle calculation, particularly in situations when onboard sensors may be inadequate or inaccurate.
- Artificial-Intelligence-Augmented (AIA) approach - To predict the sideslip angle, the artificial intelligence augmented (AIA) approach combines machine learning and AI approaches. In order to precisely estimate the sideslip angle, AIA techniques can understand intricate correlations between sensor data and vehicle dynamics by utilizing algorithms like neural networks, support vector machines, or deep learning models. AIA techniques are flexible and adaptive, enabling reliable estimation even in difficult driving situations when conventional techniques would falter.

B. Trajectory Tracking Control

Autonomous vehicles (AVs) depend on vehicle trajectory tracking control to compute appropriate directives such as steering angles, throttle opening, and braking pedal inputs to replace or support the driver. This control module makes sure the car stays clear of obstructions and drives in a safe manner by precisely following a predetermined route. The vehicle's speed, acceleration, and steering are adjusted using feedback or self-learning tracking control algorithms at the microscopic level to follow the planned trajectory, taking into consideration the kinematics and dynamics of the vehicle as well as environmental uncertainties and disturbances.

Numerous testing, such as simulation, hardware-in-the-loop, track, and on-road tests, are carried out to verify the safety and dependability of these control systems. The AV industry has seen a concentration on the development of tracking control algorithms by institutions, automakers, and component suppliers. These algorithms can be broadly divided into three categories:

- 1) feedback control without prediction (like PID, LQR, and SMC): Readjusting control inputs purely on the basis of current system state measurements is known as feedback control without prediction. This category includes algorithms such as PID, LQR, and sliding mode control.

They continuously compute control directives to reduce the error between the desired trajectory and the vehicle's current state. These control strategies are reactive in that they only take into account the system's predicted or future behavior in response to sudden deviances from the intended path.

- 2) feedback control with prediction (like MPC): Model Predictive Control (MPC), an example of feedback control with prediction, is predicting the system's future behavior based on its dynamics and current state. Typically, MPC takes into account both present and future variables and restrictions when optimizing control inputs to minimize a cost function over a given prediction horizon. Unlike approaches without prediction, MPC allows for more accurate and proactive control by dynamically adjusting the trajectory to account for predicted changes in the system or environment through iterative updates of the control inputs over time.
- 3) learning-based control (like deep reinforcement learning): By using machine learning approaches, learning-based control systems, including deep reinforcement learning (DRL), allow the vehicle to learn control policies through interaction with its surroundings. Through trial and error and input from the environment in the form of incentives or penalties, an agent in DRL learns the best control actions. As time passes, the agent refines its control strategy by optimizing cumulative rewards and modifying its actions in response to prior learning. Although there are still many obstacles to overcome in terms of training and guaranteeing safety in practical applications, DRL offers the possibility for adaptive and flexible control in complex and uncertain contexts.

C. Collaborative Control

The autonomous vehicles are having limitations in fully perceiving dynamic environments solely with onboard sensors, citing factors like adverse weather conditions and uncertainties. The integration of Internet of Things(IoT) and Artificial Intelligence(AI) in Intelligent Transportation Systems(ITS) addresses the limitations facilitating connectivity among various traffic participants and enhancing transportation efficiency, emission reduction, and accident prevention. One interesting approach in ITS is the emergence of collaborative control for connected autonomous vehicles (CAVs). Through Vehicle-to-Everything (V2X) connectivity, collaborative control enables coordinated decision-making across CAVs, enhancing economy, safety, comfort, and traffic efficiency. This method makes use of CAV cooperation to enable large-scale optimal vehicle control. [12]

VI. ELECTRICAL AND ELECTRONIC ARCHITECTURES

The automotive electrical/electronic(E/E) architectures although play a pivotal role in modern vehicles, they introduce challenges arising from increasing complexity of E/E components and software code. To tackle this complexity and reduce

wiring harnesses E/E architecture has evolved from point-to-point connections to digital-bus paradigms like Controller Area Network(CAN). Furthermore, networking technologies such as automotive Ethernet and Time Sensitive Networking have emerged to meet the demands of autonomous vehicles against the traditional bus technologies like CAN,Local Interconnect Network (LIN) and FlexRay.

A. The Current Landscape of E/E Architecture Topologies in Academic and Industrial Spheres

Scholars in the field have made significant contributions to our understanding of E/E architecture topologies by addressing various challenges and proposing novel approaches. Among these contributions are the groundbreaking studies of Stolz and Navale , which introduced roadmaps detailing the evolution of E/E architecture topologies. Other scholars, like Haas, Baic, and Saidi, built upon these roadmaps and emphasized concepts like zone orientation. Furthermore, studies by Yu, Zeng, and Mody offered insightful analyses of future trends in E/E architecture, including centralized architectures and domain-based structures.

In order to satisfy the needs of contemporary automobiles, OEMs and Tier 1 suppliers have demonstrated a strong interest in advancing E/E architecture topologies. Businesses including Tesla, Volkswagen, GM, BMW, and Toyota have experimented with novel ideas or new E/E architecture techniques. Leading vendors including BOSCH, Continental, Vector, Aptiv, and Huawei are actively engaged in E/E architecture research and development, platform customization, and the creation of essential components for new designs. All things considered, a serial roadmap for the evolution of E/E architecture, first put forth by organizations such as BOSCH, has become well known. Six phases make up this roadmap: cloud computing for vehicles, domain fusion, integration, centralization, modular, and vehicle computers with Zone Control Units.

B. Misconceptions and Queries on Serial Roadmap

Academic and industrial research and development have been significantly impacted by the current serial roadmap for the development of E/E architecture topologies. A central vehicle computer is required in zone-oriented architectures, although there are still several misconceptions about this relationship. These include the relationship between zone-oriented and domain-oriented systems.

Divergences from the serial roadmap have been observed in real-world engineering scenarios to accommodate a range of product needs and commercial goals. Unconventional topologies, such as those in the Tesla Model 3 and the communication and computer architecture of Huawei, are two examples. These differences highlight the difficulties in applying the serial roadmap to achieve universality.

Additionally, there are misconceptions among OEMs and Tier 1 suppliers as a result of the serial roadmap's inadequate handling of the evolution of zone-oriented architectures. In order to address these misunderstandings and questions, a

parallel roadmap that offers a more nuanced view of the development of E/E architectures must be taken into consideration.

C. Parallel Roadmap for Development of E/E Architecture Topologies

A parallel roadmap for the development of E/E architectural topologies is given in order to address current misconceptions and questions and meet the varied needs of OEMs and Tier 1 suppliers. The present parallel development roadmap of E/E architecture topologies, future focused and vehicle cloud E/E architectures, and classic E/E architectures comprise the three stages of this parallel roadmap.

In order to represent the state of the art and different requirements, the current roadmap divides development into two routes: zone-oriented and domain-oriented. The domain-oriented approach culminates in a single vehicle computer by means of a step-by-step integration of functions into domain control units (DCUs). On the other hand, the zone-oriented approach moves toward a more centralized design with high-performance computing cores by going through phases of zone-centralization, part-zone fusion, and full-zone fusion.

Key differences between these routes are summarized in Table 1, covering aspects such as ecological compatibility, scalability, functional safety, backbone network requirements, and harness cost savings. Overall, the trend leans towards centralized and cloud-based architectures, aiming for increased computational power, efficient sensor fusion, reduced ECU count, and lower wiring costs, while leveraging vehicle cloud computing for enhanced features and services.

D. Case Analysis of Domain-Oriented and Zone-Oriented E/E Architecture

A typical domain-oriented and zone-oriented E/E architectures establishes models focusing on their respective centralization stages. These models are based on commercial software and prioritize messages according to their relevance, assuming similar subsystem capabilities. While four zone control units (ZCUs) in the zone-oriented design connect to neighboring ECUs and either process messages internally or route them to other ZCUs, four domain control units (DCUs) in the domain-oriented architecture maintain connections with ECUs within respective domains. End-to-end worst-case transmission time (WCTT) is taken into account in network performance research as a benchmark for real-time message transmission. Radar charts are used to compare parameters including real-time capability, wiring harness arrangement, and ECU reduction.

E. Hybrid E/E Architectures

In order to achieve better overall performance, hybrid E/E designs are being developed with the goal of utilizing the benefits of both zone-based and domain-based systems. The planned road map shows the steps that this evolution will take place. In order to reduce wire complexity, the hybrid architecture initially keeps DCUs in place but transfers functions to zonal controllers. DCUs eventually combine to form

a centralized control, with high-speed Ethernet connecting them to ZCUs or other DCUs. In the end, several DCUs are replaced by a high-performance vehicle computing platform that integrates with ZCUs and connects to the cloud. By combining the advantages of zone-oriented and domain-oriented techniques, hybrid designs provide overall performance while maximizing scalability, real-time performance, and wiring harness organization for a range of message transports. [13]

VII. ROLE OF A HUMAN IN AUTONOMOUS SYSTEMS

Determining the system boundary is an issue when defining safety standards for autonomous systems (AS), especially when it comes to human involvement. The integration of humans complicates the standard systems analysis process, which entails defining what is in the environment and what is inside the system.

Theoretically, people may be seen as either part of the environment or as operators distinct from the system. But in reality, AS frequently depend on human contact for a variety of purposes, thus it's important to comprehend how humans and AS interact with the system.

The main problem is balancing the need to automate tasks within AS with the requirement for human interaction. When defining the system boundaries and determining the necessary safety measures, human elements such as decision-making, situational awareness, and response skills must be carefully taken into account. In conclusion, defining the system boundary in the context of AS necessitates addressing the intricate relationship between automation and human participation and making sure safety standards take into consideration both human aspects and technology capabilities. The possible human roles with safety implications for any Autonomous Systems can be considered as follows.

A. Human as a Function for Safety

Within the framework of autonomous systems (AS), people frequently act as safety medium by being available to step in and resolve problems. This is a role that has long been played by people supervising both other people and low-autonomy machines. But while guaranteeing the dependability of this safety function, it is often disregarded or undervalued in conversations about AS. Furthermore, political exigency or legal requirements may require human intervention in AS; this is especially true for innovative systems such as autonomous vehicles (AVs). Numerous systems in place, including the armed forces of a country and the commercial aviation sector, are predicated on the idea that a single person is always ultimately in command and accountable for making decisions. This emphasizes how crucial it is to take political and legal issues into account while integrating humans into AS frameworks.

B. Human doing something an AS cannot.

In order to gain major advantages over the present interactive, inhabited, and remote-control systems, many in the autonomous systems community strive toward high autonomy,

TABLE I

Characteristic	Domain-Oriented Route	Zone-Oriented Route
Features	Functions of ECUs are gradually integrated into DCUs, cross-domain DCUs and vehicle computer	Components of sensors, actuators and ECUs are connected to ZCUs nearby, with functions partly and fully centralized
Ecological compatibility	Roadmap is relatively smooth, which satisfies the ecology of most OEMs and Tier1 suppliers	More suitable for new car manufacturers in automotive industry (e.g., Tesla), and new Tier1 suppliers (e.g., Huawei)
Scalability and interchangeability	Subject to more limitations, such as wiring harness, intra-domain communication load, etc	Support more functional variants, plug and play with strong replaceability
Functional safety and real-time property	Integrate into domain controller according to function, it is easier to design the ASIL functional safety level of the domain controller. This task is mostly accomplished by Tier1 suppliers	Difficult for functional safety design of ZCU to be provided by traditional OEMs to Tier1 suppliers for separate design. ZCU has high requirements for functional safety and multi-core controllers
Backbone network requirements	Most of data transmission is realized in each functional domain with relatively low requirements on transmission rate and transmission characteristics	All kinds of signals will be transmitted to target electronic unit through backbone Ethernet, which imposes requirements and characteristics on transmission rate of backbone network
Harness cost saving	Less reduction in length and weight of wire harness	Reduce wiring harness and improve wiring efficiency and wiring automation

which is the ability of systems to execute general commands without human interaction or oversight. Attaining high autonomy, meanwhile, is fraught with difficulties.

Humans can be included into autonomous systems as providers of specific roles or capacities to handle jobs that cannot be automated, whether for safety reasons or to accomplish the main aim of the AS. This post may occasionally involve both mission-related and safety duties. In this role, people are integrated into the AS's mechanism and need to be treated as essential parts of the system for analysis.

C. Human as the Required General Intelligence

Because they believe that safety can only be guaranteed by clearly specified processes, some safety engineers and researchers have historically treated human operators as little more than process followers. Wright and Dekker have brought attention to new research that refutes this idea, showing that following procedures in the actual world requires a great deal of interpretation and judgment that is beyond the capacity of existing computers and robots. Furthermore, proponents of Resilience Engineering, such as Hollnagel, contend that human engagement is essential since safety in complex systems frequently depends on inventive improvisation beyond rigid processes.

Advances in autonomy concentrate on specialized capabilities rather than general intelligence, even as current suggestions for autonomous systems fall short of human-level intelligence. For certain activities, even the most sophisticated learning and planning algorithms need to be highly customized. According to Dreyfus, Searle, and Penrose, some people believe that human-level AI is unlikely to be achieved. However, these claims are still debatable and have drawn rebuttals from AI proponents like McCarthy. The fact that human thinking still requires human thinkers after decades of AI research, in spite of continuous arguments, emphasizes the complexity and unpredictability surrounding AI's future.

D. Humans as Liaison and Mediators

People are essential in the setting of autonomous systems because they serve as interpreters and communicators between

the AS and other people. This function requires both parts of translation because it entails not only interpreting information and commands but also facilitating control or collaboration between humans and AS. In tactics where human translation is crucial, such as the unmanned aerial vehicle (UAV) strategy that the Civil Aviation Authority has embraced, this position is especially important. Because intelligent communication is nuanced, human comprehension and mental model updating are still essential, even with the most advanced voice synthesis and interpretation technologies.

E. Human as Maintainer

Autonomous systems (AS) will become increasingly dependent on human involvement for maintenance and support in the near future as they continue to develop. This includes a number of crucial tasks, such as data management, software upgrades, hardware upkeep, system monitoring, and emergency response. To guarantee dependability and performance, humans are required for the development, testing, and implementation of software updates as well as for the inspection, replacement, and repair of physical components. In addition, human intervention is essential for data management and curation, system performance monitoring, problem diagnosis, emergency response, and maintaining the security and effectiveness of AS operations.

The complexity and unpredictability of maintenance activities mean that human knowledge and oversight will always be necessary, even though advances in robots and AI may ultimately allow AS to do some maintenance chores autonomously. Thus, for the foreseeable future, human intervention will still be necessary for the autonomous systems to function reliably.

F. Human as Peer

Predictability of behavior becomes vital in human-autonomous system (AS) interactions, especially in safety-sensitive circumstances where they cooperate and share resources. This means that AS behavior needs to be constrained in ways other than fulfilling their main mission or preventing serious mishaps. While interoperability constraints may limit

potentially safe individual activities, they also guarantee overall system safety, as seen by examples from laws such as the Highway Code and the UK Rules of the Air. For example, the UK's mandate that cars drive on the left makes it easier to manage multiple vehicles at once safely. Therefore, in order to support humans as peers, AS behavior may need to be constrained in ways that compromise some performance and flexibility in favor of safety and predictability. The delicate balance between optimization and risk mitigation in AS design and operation is highlighted by the possibility that an AS is sometimes too intelligent and adaptable to guarantee safety. [14]

VIII. KEY TECHNOLOGIES

This section explores the primary tasks that autonomous vehicles (AVs) carry out, such as course planning, obstacle avoidance, and traffic sign detection, all of which reduce the workload for humans. It looks at the developments in planning, control, and perception technologies that make it possible for AVs to carry out these jobs efficiently and simultaneously.

A. Environment Perception

The crucial function of environmental perception in autonomous vehicles (AVs) is described in this paragraph, which also covers a variety of sensor systems, including radar, laser, and visual navigation. It draws attention to the difficulties with object detection, such as partial views, occlusions, and changes in the weather and illumination. Accurate object detection in real-time from noisy LiDAR point clouds depends on deep learning (DL) techniques, especially YOLOv3. Deep learning (DL) is used for a wide range of applications, including object discrimination, vehicle tracking, pedestrian recognition, and trajectory prediction using RGB, LiDAR, and RADAR sensor data.

B. Pedestrian Detection

In autonomous cars, pedestrian detection is the process of identifying pedestrians with the use of sensors. It includes segmentation, feature extraction, segment classification, and track classification. There are issues with hazy weather, which makes it difficult to see and results in hazy outlines in photos taken with a camera. In order to overcome this, techniques based on 3D LiDAR data have been put forth; these involve condensing LiDAR data into 2D images and using CNNs to precise detection. Additionally, less expensive algorithms have been created that use sub-networks for pedestrian recognition and regression and are based on 2D LiDAR and monocular pictures. Process tracking and segment categorization together can split point clouds into separate groups and combine active characteristics for classification, hence resolving occlusion problems. Track classification improves pedestrian recognition even more.

C. Path Planning

Autonomous vehicles use onboard sensors such as LIDAR, cameras, GPS, radar, and inertial sensors to discover the best

routes between two sites in real-time. This process is called trajectory planning, or path planning. Numerous path planning methods have been created, including the frenet coordinate system and deep learning-based lane recognition algorithms. Discrete optimization techniques have also been developed. These algorithms compute several paths and select the least expensive one; they frequently employ cubic spline interpolation for smoother paths. For 3D SLAM, multisensor fusion systems that include GNSS, LiDAR, and INS are suggested in order to increase accuracy and resilience.

These systems offer great accuracy even in difficult settings, making up for shortcomings in earlier versions. The goal of integrating these technologies is to satisfy automated driving systems' strict accuracy standards. These systems can manage a variety of scenarios, such as deep urban regions and extended GNSS outages. All things considered, these developments lead to safer and more effective autonomous driving capabilities.

D. Vehicle Cyber Security

The automobile industry's progress in autonomous driving technology has resulted in the extensive integration of various components such as actuators, communication systems, and sensors in cars. But this added complexity also means that there are more ways for cyberattacks to occur, giving attackers access to vehicles from the outside. Numerous security-related initiatives have been launched by governments, automakers, and IT businesses globally in order to address this risk. In order to give testers and security engineers a consistent description of cyberattacks, efforts to categorize and analyze cyberattacks must be coordinated.

For example, a taxonomy for categorizing security threats has been created, offering important information for future study. Another area that needs exploration is remote control of autonomous cars, especially in situations where vehicles are attacked. In these situations, drivers could need to take over management of the operation; however, completely autonomous cars might not be able to implement this strategy. It is suggested to use Internet of Things (IoT) sensors in Intelligent Transportation Systems (ITS) to solve issues with security and remote vehicle control. To improve cybersecurity in autonomous cars, other strategies include separating attack signals and using jammers to stop intruders without interfering with communication.

E. Motion Control

Autonomous vehicles (AVs) rely heavily on motion planning algorithms to find safe and effective routes in a variety of situations. To guarantee safe interactions, these algorithms also need to take neighboring vehicle driving habits into account. Dang et al. delineate three fundamental classifications of motion control algorithms: Sampling-based techniques, optimum control techniques, and Artificial Potential Field. By determining the shortest path along the gradient, the Artificial Potential Field approach creates motion fields based on target attraction and obstacle repulsion. It might not, however, take future driving behavior into consideration.

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IX. BLOCKCHAIN SOLUTIONS FOR AVS

Autonomous vehicles (AVs) have attracted a lot of interest in academia and industry for a variety of uses, including home assistants, driverless automobiles, automated warehouses, and drone delivery systems. However, because of the intricacy of antivirus software and its subsystems, there are weaknesses that can be taken advantage of by unethical activities, such as data breaches and cyberattacks. Blockchain technology is gaining traction as a strong answer to these issues because of its security features, which include immutability, transparency, and decentralization.

Blockchain technology offers several advantages over traditional security mechanisms:

- Decentralization: Highly resilient to external attacks.
- Data Integrity: Since blockchain is immutable and transparent.
- Transparency and Trust: Provides equal access to stored data and requires consensus for any change.
- Immutable Records: Creates a permanent and tamper-proof record of transactions.
- Privacy and Security: Contains strong cryptographic primitives and privacy management techniques.

Proof of work (PoW) is one of the consensus processes used by blockchain technology to thwart malicious access and tamper-proof the data. AVs can guarantee private communication, lessen vulnerabilities, and improve system security by utilizing blockchain. To better secure data related to diverse AV operations, numerous distinct distributed ledgers can be made for various stakeholders. This would ultimately reduce errors and on-road accidents.

Blockchain technology is quickly moving from a theoretical idea to an actual product, transforming a number of industries like food supply chains and healthcare. Blockchain provides a decentralized network that guarantees safe, confidential, and transparent transactions in the age of autonomous vehicles (AVs), where millions of IoT devices must communicate and exchange data instantaneously. Automakers hope to use blockchain technology to advance autonomous vehicles (AVs), boost consumer happiness, and simplify a number of vehicle management processes.

Through the use of blockchain technology and smart contracts, it is being investigated how digital identities, such as vehicle identification, ownership, warranties, leasing, parts management, and servicing data, might be maintained in a shared ledger structure. According to their needs and use permissions, authorized parties will have access to particular data, enabling certificate usage and authentication in the near future.

Every user has a copy of the data in a distributed record-keeping system called blockchain, which is safe and unchangeable. Updates may only take place with the consent of all users. This permits decentralized transactions, lowers costs and risks, and guarantees data integrity. Blockchain-based systems are robust because of techniques like authorization mechanisms, distributed ledgers, consensus algorithms for synchronization, and transaction encoding as blocks. Blockchain is essential to the AV architecture network because it connects cars and IoT sensors, enabling control, monitoring, and guidance based on service requirements. Peer nodes make up the blockchain network, whereas miner nodes verify trust. Blockchain technology makes it possible to follow and record the activity of vehicles, authorize authorized users, and quickly identify and address assaults or compromises of IoT sensors. Overall, blockchain enhances security, transparency, and reliability in AV operations. [16]

X. EVALUATION OF EXCLUSIVE LANES FOR AVS

An extensive simulation was carried out as part of the assessment of AV-exclusive lanes on Seoul, South Korea's expressways in order to determine any possible effects on traffic flow and network performance. The study's main conclusions and suggestions are outlined below:

Overall Improvement in Network Performance: Based on the simulation findings, it was found that using AV operating scenarios improved network speed and traffic volume overall, especially when the mixed traffic percentage (MPR) rose.

Impact Variability: On the other hand, AV lanes' effects differed according on service locations and traffic capacity, among other things. The Olympic Expressway unexpectedly observed drops in both vehicle kilometers traveled (VKT) and total travel time (TTS) despite increased traffic volume, whilst the Gyeongbu Expressway saw increases in VKT and declines in TTS. This was ascribed to substantial changes in travel routes caused by increases in traffic flow, mostly on shorter links close to important business sectors.

Concerns about Efficiency at Low MPR: The study drew attention to inefficiencies that could arise from underutilization of AV-exclusive lanes, which could cause traffic congestion. The study found that for expressways with AV-exclusive lanes to function efficiently, an MPR of 20–30

Limitations and Future Directions: The research admitted that its assumptions about travel behavior and AV operating circumstances were not entirely accurate. Future studies should take into account more realistic AV operating conditions and travel behavior changes brought about by the use of AVs. Furthermore, microscopic analysis is required to record individual

vehicle behavior and evaluate the environmental impact of AV-exclusive lanes, especially in light of the possibility that AVs be battery-electric vehicles.

In conclusion, even though AV-exclusive lanes have the potential to increase urban mobility, their deployment will only be successful if key variables including traffic patterns, operational effectiveness, and environmental impact are carefully taken into account. [17]

XI. CONCLUSION

The writings address a variety of subjects pertaining to autonomous systems, such as the development of automotive electrical/electronic (E/E) architectures, autonomous vehicles (AVs), and intelligent autonomous agents (IAAs). This is an overview:

Perception of Technical Autonomy: The study investigates how social perception is impacted by increasing degrees of autonomy in technological systems, like self-driving automobiles and predictive police software. It implies that technological functionality, especially autonomy, has an impact on how people perceive technology in society and plays a role in human-machine relations.

Success Factors and Inhibitors of Autonomous Vehicles: The significance of reliable sensor data, security concerns, and the function of wireless technologies for cooperative driving are only a few of the important success factors and inhibitors of autonomous vehicles that are covered in this study. It addresses the difficulties and weaknesses related to sensor and camera security and highlights the requirement for secure wireless technology to ensure effective implementation.

The paper discusses Intelligent Autonomous Agents (IAAs) in general, emphasizing their potential for high-level cognitive tasks and the integration of current AI techniques. It addresses problems with communication, negotiation, coordination, conflict, and social behavior among autonomous agents, as well as future research goals and difficulties.

Trustworthiness of Autonomous Systems: The chapter looks at how important it is for autonomous systems to be trustworthy. It makes the case that sensible, logical, and ethical standards must be followed by robots and autonomous systems in order for them to be effective. It talks about what makes people trust autonomous systems and stresses that interdisciplinary research is necessary to create reliable autonomous systems.

The evolution of automobile electrical/electronic is examined in this article, which also suggests parallel roadmaps for zone-oriented and domain-oriented designs. It talks about how networking technologies like Service-Oriented Architecture (SOA), Software-Defined Networking (SDN), and Time-Sensitive Networking (TSN) are applied in E/E systems and looks ahead to integrated networking technologies in the future.

Last but not least, the book discusses both theoretical and practical issues pertaining to autonomous systems, such as the necessity of limits and the issue of human involvement. The text proposes a practical methodology grounded in the principles of complex adaptive systems (CAS) and underscores

the significance of thorough modeling and analysis in tackling issues related to autonomous systems.

In general, the writings offer valuable perspectives on a range of topics related to autonomous systems, such as their social perception, success factors, technical difficulties, reliability, architectural development, and pragmatic concerns.

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